

Observe & Debug

Timescale

My initial observation (and debugging) of the interaction between the timescale system and the effect on the character movement component was that UE5 does not modify the velocity variables directly and instead properly actually scales the physics simulation.

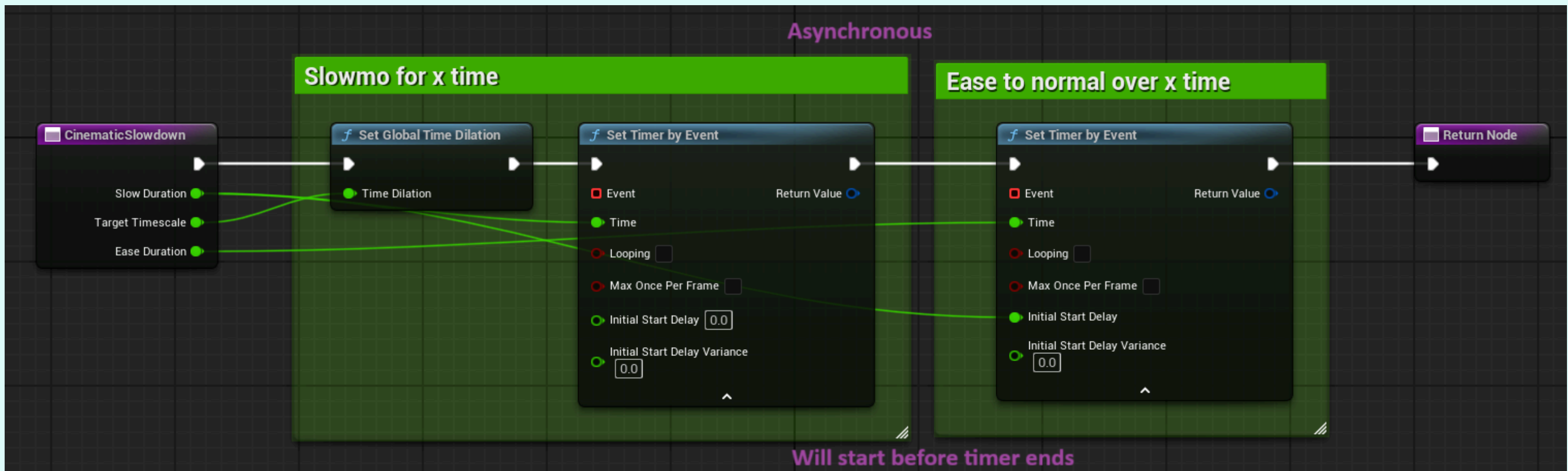


My process was to look into how timescale works in blueprint vs timescale via the console command “slomo”. Under the hood they modify the exact same value, so I was able to just set the timescale to 0.5 and do the exact same action and then compare the results of each.

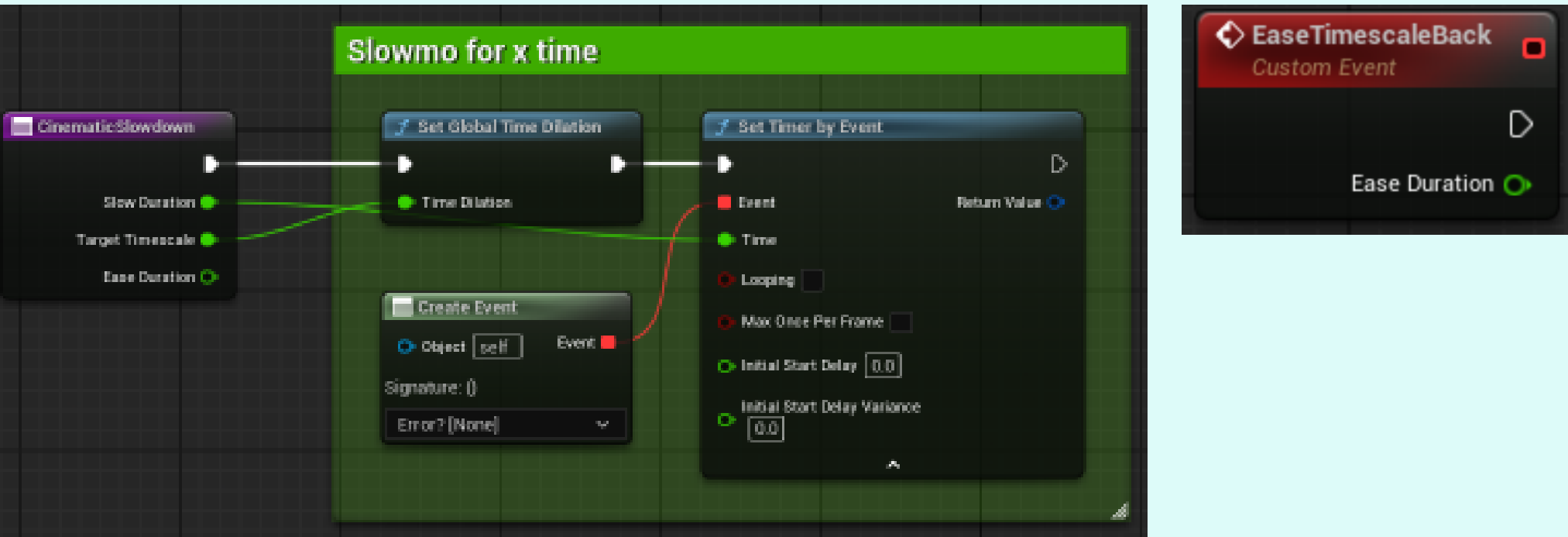
Modification

Slowmo during explosion

Initially when I looked into trying to create a function to slowdown for X amount of time then to ease back in, I ran into a major problem which was due to my BP being in a function that would be called I can’t actually make it Synchronous



So I tried using an event since that could be called elsewhere and function Synchronously by calling other events based on timers. However this doesn’t work as I need to pass the ease duration through, which is impossible for some reason here

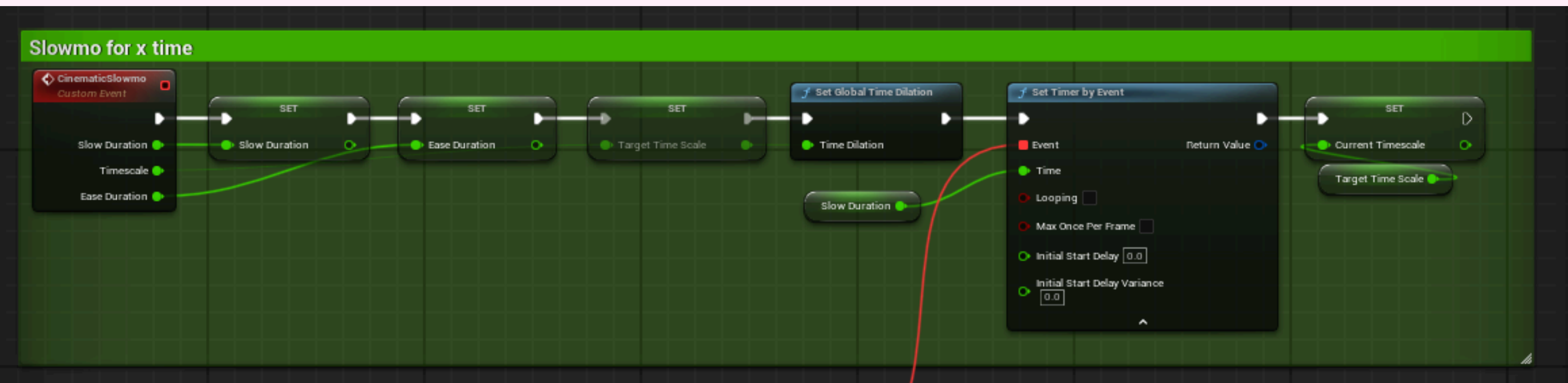


Additional Observations + Debugging

Whilst debugging this I ran into an excessive amount of issues due to various sources of jank.

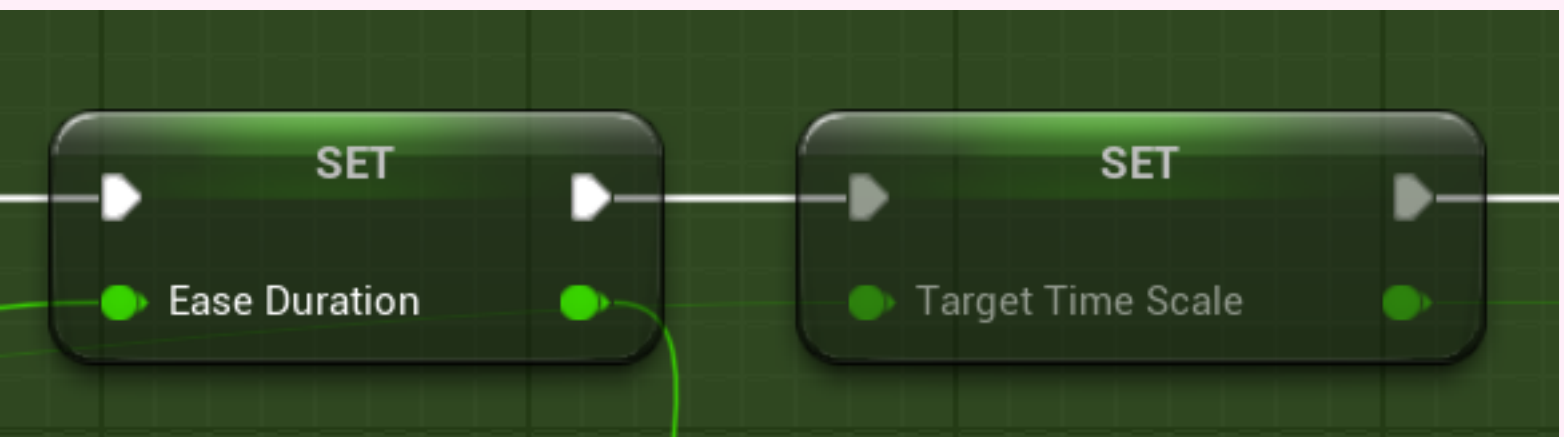
For example, this seemed to not be able to properly set the CurrentTimescale to the value it had read.

Can you spot the issue?



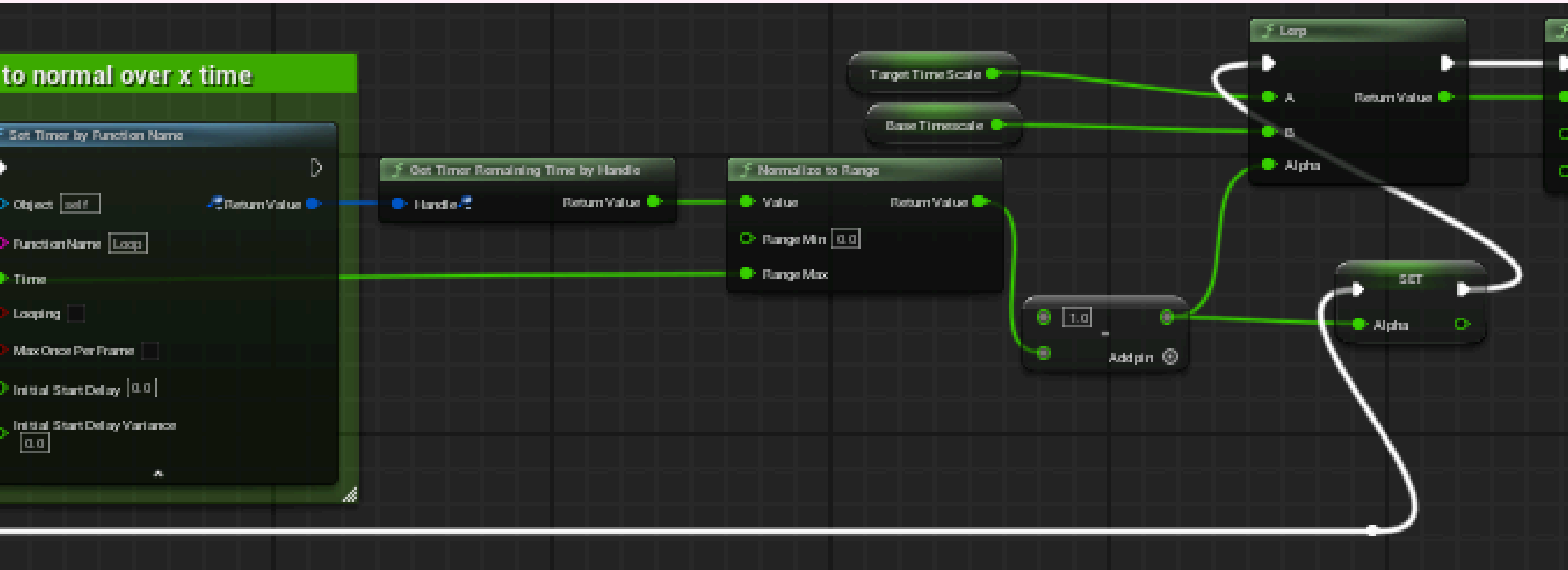
Looks normal right? wrong!

Turns out the only difference in a node being enabled or disabled is a very slight tint of transparency on the variable name. If you zoom in enough you can start to see the slight difference which is only apparent at a high zoom level



The one on the right is disabled

After debugging that for an hour I went onto the next issue which I had tried a different approach



This time I tried ensuring the alpha was updated constantly and as shown below I was using a boolean to track the state of if we were easing. this did not seem to work and still despite what I had done just set the timescale to 1 seemingly instantly.

